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ORIGINAL ARTICLE

The utility of quantitative calf muscle near-infrared spectroscopy in the follow-up of acute deep vein thrombosis

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Summary. *Background:* To investigate patterns of venous insufficiency and changes in calf muscle deoxygenated hemoglobin (HHb) levels after an acute deep vein thrombosis (DVT). *Methods:* A total of 78 limbs with an acute DVT involving 156 anatomic segments were evaluated with duplex scanning and near-infrared spectroscopy (NIRS) at 1, 3, 6 and 12 months. Venous segments were examined whether they were occluded, partially recanalized, and totally recanalized, and the development of venous reflux was noted. The NIRS was used to measure calf muscle HHb levels. Calf venous blood filling index (HHbFI) was calculated on standing, then the calf venous ejection index (HHbEI), and the venous retention index (HHbRI) were obtained after exercise. *Results:* The segments investigated were the common femoral vein (CFV; 38 segments), femoral vein (FV; 37), popliteal vein (POPV; 44), and calf veins (CV; 37). At 1 year, thrombi had fully resolved in 67% of the segments, 27% remained partially recanalized, 6% were occluded. The venous occlusion was predominant in the FV (24%) at 1 year. On the contrary, rapid recanalization was obtained in CV than proximal veins at each examination ($P < 0.01$). Venous reflux was predominant in POPV (55%), followed by FV (19%), and no reflux was found in CV. At 1 year, the HHbFI in POPV reflux patients was significantly higher than those with resolution (0.19 ± 0.14, 0.11 ± 0.05 μM s, $P = 0.009$, respectively). Similarly, there was a significant difference in the HHbRI between the two groups (3.08 ± 1.91, 1.42 ± 1.56, $P = 0.002$, respectively). In patients with FV occlusion, the value of HHbRI was significantly higher than those with complete resolution (2.59 ± 1.50, 1.42 ± 1.56, $P = 0.011$, respectively). *Conclusions:* The lower extremity venous segments show different proportions of occlusion, partial recanalization, and total recanalization. The CV shows more rapid recanalization than proximal veins. The NIRS-derived HHbFI and HHbRI could be promising parameters as the overall venous function in the follow-up of acute DVT. These findings might be very helpful for physician in detecting patients who require much longer follow-up studies.

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Introduction

Post-thrombotic syndrome (PTS) is considered to be a common late complication of an acute deep vein thrombosis (DVT). Severe PTS might be caused by gravitational venous reflux, venous obstruction, calf muscle pump malfunction, or combination of these [1–6]. The PTS, however, is difficult to predict in acute phase of DVT. Individual anatomic venous segments in the lower extremity respond in a different way to the presence of thrombus [7]. In the late phase, the presence of reflux in the femoral and popliteal vein (POPV) is considered to contribute the development of PTS, but the role of an obstruction, reflux, or a combination of the two in the rest of venous segment remains unclear [8].

Near-infrared spectroscopy (NIRS) is a non-invasive optical method to determine oxygenation and hemodynamics in the tissue. Using a modified Beer–Lambert law [9], it is possible to obtain a quantitative value for total hemoglobin (tHb), oxygenated hemoglobin (O_2Hb), and deoxygenated hemoglobin (HHb) levels. Recent studies have shown the validity of NIRS in non-invasive measurement of tissue oxygenation in peripheral arterial disease [10–13]. These reports grade the severity of peripheral arterial disease by assessing resaturation of the calf muscle after exercise. On the other hand, Hosoi *et al*. investigated the severity of chronic venous insufficiency (CVI) by measuring the changes of HHb and concluded that NIRS is possible useful in assessing the ambulatory venous function of primary valvular insufficiency as well as PTS [14,15].

In this study, venous abnormalities including compressibility and reflux of the lower extremity venous system were evaluated with duplex scans, and the patterns of calf muscle HHb levels were measured using NIRS. The purpose of this study was to investigate the correlation between the findings obtained by

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duplex scans and calf muscle NIRS, and to determine the possible role of NIRS in the follow-up of DVT.

Materials and methods

Patients

From January 2002 to September 2003, a total of 91 limbs in 88 patients with duplex-confirmed acute DVT were prospectively evaluated with follow-up duplex scanning and NIRS. After initial diagnosis, all patients were treated with i.v. unfractionated heparin for 5–14 days during the acute phase, followed by oral warfarin for at least 8 weeks with International Normalized Ratio (INR) level of 2–3. All patients were encouraged to ambulation, and graduated thigh-high compression stockings were applied. The patients in this study were placed on follow-up at 1, 3, 6 and 12 months. At each visit, patients were asked how compression stockings fitted. Informed consent for anticoagulation therapy and follow-up by duplex scanning and NIRS was obtained by all subjects. The exclusion criteria included patients who did not fully ambulate with the walking distance of < 500 m because of continuing medical problems, and patients with arterial insufficiency, identified on the basis of an ankle-brachial pressure index of < 0.9. Patients with poor compliance of the use of compression stockings were also excluded from the study.

Venous duplex scans

The presence of DVT was diagnosed with duplex scans. A color duplex scanner (LOGIQ 500MD: GE Medical Systems, Milwaukee, WI, USA) with a 5–10 MHz transducer was used. Initially, each patient was placed supine in a reverse Trendelenburg position at 15°. Venous duplex scanning began at the common femoral vein (CFV), and moved to the femoral vein (FV) at the adductor canal. The anterior and posterior tibial veins (ATV and PTV) were also recorded. Afterwards, the patient was placed in a prone position with the knee flexed at 30°, and the residual popliteal (POPV), peroneal (PV), gastrocnemius (GV), and soleal veins (SV) were evaluated [16]. The diagnosis of DVT was based on both non-compressibility of the vein on B-mode, and no spontaneous flow on color Doppler imaging. Thrombosis was considered as proximal if thrombus was detected in CFV, FV, and POPV with or without calf vein (CV) thrombosis. Thrombosis was considered as distal if thrombus was detected only in the CV. Duplex scanning was repeated at 1, 3, 6 and 12 months. At each examination, venous segments were assessed to determine the distribution of thrombosis and whether they were occluded, partially recanalized, and totally recanalized. Venous segments were considered as occluded if they showed incompressibility and absence of flow with distal augmentation. Partial recanalization was defined by partial compressibility of the vein and diminished cephalad flow by distal augmentation. Complete recanalization was determined by complete compressibility of the vein and spontaneous cephalad flow on color Doppler imaging.

The development of venous reflux was evaluated as described by van Bemmelen *et al*. [17]. Valve closure times (VCTs) were measured in the deep venous systems, with the patient in the upright position with his/her weight supported on the contralateral leg. For evaluation of the CFV and the FV, a pneumatic thigh cuff (Hokanson, Bellevue, WA, USA) was attached to the thigh, inflated to 80 mmHg and then rapidly deflated. For evaluation of the POPV, a cuff was applied to the calf, inflated to 100 mmHg and then rapidly deflated. Calf veins were examined with the patients in the sitting position, with the foot resting on a stool, distal calf compression was undertaken. VCTs of > 0.5 s were considered to be incompetent.

Near-infrared spectroscopy

The calf muscle HHb levels were obtained by a NIRS (OM-200, Shimadzu Co., Kyoto, Japan). The NIRS technique is well-established, non-invasive and continuous, and operates in real time, leading to *in vivo* monitoring of tissue oxygenation and hemodynamics during certain physiologic process [18–20]. Briefly, with this method, optimal absorption is measured across tissue at four different wavelengths: 690, 780, 805 and 830 nm, and changes in these parameters are calculated from the experimental optical density values. This instrument was used with a sensor with a 3.5 cm distance between light source and detector. The sensor was placed on the posterior aspect of the calf firmly with adhesive tapes. Changes in the hemoglobin concentration were expressed as μM.

To assess the ambulatory venous function, measurements were carried out using the standard exercise protocol for air plethysmography (APG) [21]. The protocol began with the patients resting in the supine position with the leg elevated on a foam block for 5 min. Increases in HHb were observed in the recording when the patients were instructed to a standing position without putting any weight on the leg being studied. The patients were asked to keep still until a plateau (HHbV) was reached. Calf venous blood filling index (HHbFI) was calculated by dividing 90% of the venous blood volume by the time it takes to fill 90% of the venous volume. Then the patients were asked to stand on tiptoe with the weight on both legs, which produced an ejected volume (HHbEV), and return to the initial position, and the changes in the HHb were observed. The calf venous ejection index (HHbEI) was calculated by $\mathrm{HHbEI} = \mathrm{HHbEV}/\mathrm{HHbV}$. After a new plateau is reached, the patients were asked to do 10 tiptoe movements, leaving a venous expulsion (HHbE) and a subsequent retention (HHbR). The venous retention index (HHbRI) was determined by $\mathrm{HHbRI} = \mathrm{HHbR}/\mathrm{HHbE}$ (Fig. 1).

Statistical analysis

All data were analyzed using STATVIEW for Windows (Version 5.0, SAS Institute Inc., Cary, NC, USA). Wilcoxon's non-parametric rank sum test was used to estimate differences between numerical data, and chi-squared contingency table

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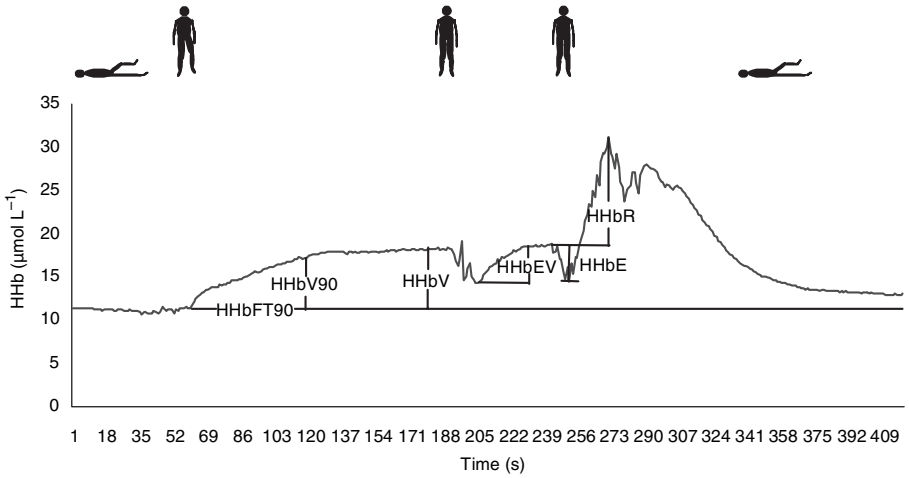


Fig. 1. The near-infrared spectroscopy (NIRS) examination: calf venous blood filling index (HHbFI) was calculated by dividing 90% of the venous blood volume by the time it takes to fill 90% of the venous volume. Then the patients were asked to do stand on tiptoe with the weight on both legs, which produced an ejected volume (HHbEV), and return to the initial position, and the changes in the deoxygenated hemoglobin (HHb) were observed. The calf venous ejection index (HHbEI) was calculated by HHbEI = HHbEV/HHbV. After a new plateau is reached, the patients were asked to do 10 tiptoe movements, leaving a venous expulsion (HHbE) and a subsequent retention (HHbR). The venous retention index (HHbRI) was determined by HHbRI = HHbR/HHbE.

analysis was used to evaluate differences between proportions. Statistical significance was defined as $P < 0.05$.

Results

At 1 year, five patients were lost to follow-up, eight had inadequate follow-up study, and the remaining 78 limbs in 75 patients, involving 156 anatomic segments, were presented in this study. Table 1 shows baseline characteristics of the 75 study patients. Among the cases, 29 patients were male and 46 were female. Patients age ranged from 21 to 86 years (mean: 59 years). DVT was most common after operation or trauma (25 patients, 33%), followed by active cancer (11 patients, 15%). Previous DVT was less common in this study (five patients, 7%). Protein C deficiency was the most common thrombophila (five patients, 7%) followed by protein S deficiency (four patients, 5%), AT III deficiency (three patients, 4%), positive antiphospholipid syndrome (two patients, 3%), and hyperhomocysteinemia (two patients, 3%). The mean duration of the use of oral warfarin was 8.6 months with the mean PT–INR of 2.2. The segments investigated were the CFV (38 segments), FV (37 segments), POPV (44 segments), and isolated CV (37 segments).

The proportions of occlusion, partial recanalization, and total recanalization in each venous segment are shown in Table 2. At 1 year, thrombi had fully resolved in 67% of the segments, 27% remained partially recanalized, 6% were occluded. Of these, the CV had the highest rate of recanalization (100%), whereas FV showed the highest rate of occlusion (24%). No occlusion was found in CFV, POPV, and CV.

Table 1 Baseline clinical characteristic of the 75 study patients

Parameters	$n = 75$ patients
Characteristics	
Mean age (years)	58.9 ± 17.2
Gender (%; male)	29 (32.6)
Duration of the use of warfarin (months)	8.6 ± 6.4
PT–INR	2.2 ± 0.2
Risk factors (%)	
Immobilization	8 (10.7)
Known malignancy	11 (14.7)
Stroke	6 (8)
Operation or trauma	25 (33.3)
Previous history of DVT	5 (6.7)
Inflammatory bowel disease	2 (2.7)
Hormon replacement therapy	3 (4)
Thrombophilia (%)	
Protein S deficiency	4 (5.3)
Protein C deficiency	5 (6.7)
AT III deficiency	3 (4)
Positive antiphospholipid syndrome	2 (2.7)
Hyperhomocysteinemia	2 (2.7)
Abnormal plasminogen	1 (1.3)

INR, International Normalized Ratio; PT, post-thrombotic; DVT, deep vein thrombosis; AT, antithrombin.

Table 2 The outcome at 1 year

Site	Number of segments	Resolved	Partial	Occluded
CFV	38	21 (55.3)	17 (44.7)	0 (0)
FV	37	11 (29.8)	17 (45.9)	9 (24.3)
POPV	44	36 (81.8)	8 (18.2)	0 (0)
CV	37	37 (100)	0 (0)	0 (0)
Total	156	105 (67.3)	42 (26.9)	9 (5.8)

Values are expressed as *n* (%).
CFV, common femoral vein; FV, femoral vein; POPV, popliteal vein; CV, calf vein.

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Table 3 The duration of time (months) for complete recanalization

Site	Number of segments	1	3	6	12
CFV	38	6 (15.8)*	15 (39.5)*	21 (55.3)*	21 (55.3)*
FV	37	3 (8.1)*	8 (21.6)*	11 (29.8)*	11 (29.8)*
POPV	44	13 (29.5)**	24 (54.5)*	35 (79.5)**	36 (81.8)**
CV	37	23 (62.2)	33 (89.2)	37 (100)	37 (100)

*Proximal veins vs. calf veins: $P < 0.001$; **Proximal veins vs. calf veins: $P < 0.01$.
Values are expressed as *n* (%).
CFV, common femoral vein; FV, femoral vein; POPV, popliteal vein; CV, calf vein.

The duration of time for complete recanalization is shown in Table 3. At 1 month, 23 of 37 segments (62%) in the CV were fully resolved. On the other hand, proximal veins had statistically lower proportions of complete recanalization: 16% of CFV, 8% of FV, and 30% of POPV (CFV and FV vs. CV, $P < 0.001$; POPV vs. CV, $P < 0.01$). At 1 year, total resolution was found in 55% of CFV, 30% of FV, 82% of POPV, and 100% of CV, and similar statistically significant differences were found between proximal veins and CV (CFV and FV vs. CV, $P < 0.001$; POPV vs. CV, $P < 0.01$).

The venous valvular competency for each venous segment is shown in Table 4. At 1 year, POPV had the highest frequency of venous reflux (55%), followed by FV (19%), whereas no venous incompetence was found in the CV.

Table 5 shows the NIRS-derived parameters between patients with POPV reflux and these with complete resolution

Table 4 The venous valvular competency for each venous segment

	Reflux (+)	Reflux (−)	Total
CFV	4 (10.5)	34 (89.5)	38
FV	7 (18.9)	30 (81.1)	37
POPV	24 (54.5)	20 (45.5)	44
CV	0 (0)	37 (100)	37

Values are expressed as *n* (%).
CFV, common femoral vein; FV, femoral vein; POPV, popliteal vein; CV, calf vein.

Table 5 The NIRS-derived parameters between patients with POPV reflux and these with complete resolution with no reflux

	POPV reflux	Resolved with no reflux	*P*-value*
On standing			
HHbFI (μM s)	0.19 ± 0.14	0.11 ± 0.05	0.009
During exercise			
HHbEI	0.11 ± 0.05	0.41 ± 0.20	0.231
HHbRI	3.08 ± 1.91	1.42 ± 1.56	0.002

*Wilcoxon rank sum test for non-parametric data.
Values are expressed as mean ± SD.
NIRS, near-infrared spectroscopy; POPV, popliteal vein; HHbFI, calf venous blood filling index; HHbEI, calf venous ejection index; HHbRI, venous retention index.

Table 6 The NIRS-derived parameters between patients with FV occlusion and these with complete resolution with no reflux

	FV occlusion	Resolved with no reflux	*P*-value*
On standing			
HHbFI (μM s)	0.08 ± 0.05	0.11 ± 0.05	0.594
During exercise			
HHbEI	0.39 ± 0.29	0.41 ± 0.20	0.906
HHbRI	2.59 ± 1.50	1.42 ± 1.56	0.011

*Wilcoxon rank sum test for non-parametric data.
Values are expressed as mean ± SD.
NIRS, near-infrared spectroscopy; FV, femoral vein; HHbFI, calf venous blood filling index; HHbEI, calf venous ejection index; HHbRI, venous retention index.

with no reflux. The HHbFI in POPV reflux patients was significantly higher than those with resolution (0.19 ± 0.14, 0.11 ± 0.05 μM s, $P = 0.009$, respectively). Similarly, there was a significant difference in the HHbRI between the two groups (3.08 ± 1.91, 1.42 ± 1.56, $P = 0.002$, respectively). The value of HHbEI in complete resolution was higher than POPV reflux, but this was not statistically significant (0.11 ± 0.05, 0.41 ± 0.20, $P = 0.231$, respectively).

Table 6 shows the NIRS-derived parameters between patients with FV occlusion and these with complete resolution with no reflux. There was no statistically significant difference in the HHbFI between the two groups (0.08 ± 0.05, 0.11 ± 0.05 μM s, $P = 0.594$, respectively). Similarly, there was no significant difference in the value of HHbEI between the two groups (0.39 ± 0.29, 0.41 ± 0.20, $P = 0.231$, respectively). In patients with FV occlusion, the value of HHbRI was significantly higher than those with complete resolution (2.59 ± 1.50, 1.42 ± 1.56, $P = 0.011$, respectively).

Discussion

Duplex scanning is a useful tool for both identifying an acute DVT and following up studies to detect subsequent venous abnormalities [2–8,22]. Caprini *et al*. studied the rate of normalization within 6 months after the diagnosis of acute DVT in various venous segments, and found that the rates of resolution of DVT were 78% in CFV, 70% in FV, 75% in POPV, and 70% in CV, and concluded that the rate of resolution were similar for the different veins [23]. On the contrary, O'Shaughnessy and FitzGerald [7] described that individual anatomic venous segments in the lower extremity respond in a different way to the presence of thrombus. A 1-year follow-up study revealed that the rates of resolution were 78% in CFV, 42% in FV, and 54% in POPV [8]. McDonald *et al*. investigated the natural history of isolated CV thrombosis, and found that 84% of the segments remained stable or resolved during the 3-month follow-up [24]. These data are coincide with our results.

Previous studies have shown that the time to complete recanalization is considered to be important in the determination of the development of venous valvular reflux [8,25,26].

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O'Shaughnessy and FitzGerald [7] found that the venous segments that resolved within 6 months had a higher competence rate than those resolved from 6 months to 1 year [8].

Severe PTS could be caused by either obstruction or reflux in the axial deep venous system, or combination of the two. Johnson *et al*. [2] found that only 3% of limbs with PTS were normal, 18% had reflux alone, 15% had obstruction alone, and 65% had both reflux and obstruction, and concluded that a combination of residual reflux and residual partial or complete obstruction in the major deep veins is more likely to be associated with the development of the PTS [25]. On the contrary, Killewich *et al*. showed that reflux in the deep veins was the main contributor to the severity of PTS [27]. Haenen *et al*. also found that reflux in the deep proximal veins significantly contributed to the development of PTS [28]. Other investigators revealed the primary importance of superficial venous reflux in the development of symptoms of PTS [29,30]. If the perforating veins become insufficient, the venous pressure in the superficial veins might increase and cause an overload of the superficial system, resulting in a gradually developing insufficiency of the superficial veins [26]. Therefore, much longer investigation will be required on this matter.

Another factor that affects the development of PTS is calf muscle pump malfunction. Although duplex scanning is a useful tool for identifying anatomic distribution of venous abnormalities, the overall venous function is not determined. Recently, APG is used as a reliable venous function test, and it provides useful information on venous reflux, calf muscle pump function, and ambulatory venous pressure in patients with CVI [21,31]. In the presence of proximal venous occlusion, however, venous filling index obtained by the APG underestimates the magnitude of venous reflux [32]. Furthermore, there was a poor correlation between the residual volume fraction and the ambulatory venous pressure when residual obstruction is present [33]. The VFI, EF, and RVF should be mainly a function of venous valvular incompetence in the absence of significant venous obstruction [34]. Therefore, parameters obtained by the APG does not always reflect the specific hemodynamic situation, and objective assessment of calf muscle pump function is still difficult in patients with acute DVT.

Near-infrared spectroscopy is useful for measuring oxygenation and hemodynamics in the tissue during rest and broad range of exercise. Although, the measurement depth is restricted to the optrode set-up, NIRS is able to obtain signals in muscle tissue. The common parameters used by the NIRS are tHb, O_2Hb, and HHb. Changes in the concentrations of these chromophores can be measured from changes in the absorbance of light transmitted into tissues using a modified Beer–Lambert law as follows [35]:

$$A = P \times E \times C + G,$$

where A = absorbance, P = optimal path length, E = the extinction coefficient of chromophore for the wavelength, C = the concentration of the chromophore, and G = attenuation of light because of background absorbance. G varies with the type of tissue examined and geometry of the optrode positioning. Assuming that G remains constant as P and E are constant, changes in the absorbance of light should be due to changes in chromophore concentration alone [35]:

$$\Delta A = P \times E \times \Delta C$$

The first observations of tissue oxygenation using NIRS were made by Jöbsis [36]. This technique has been subsequently refined and applied to the study of cerebral hemodynamics [37–42]. By choosing an accurate distance between source and detector, NIRS can measure directly in the muscle of interest, and over the last two decades, many groups have applied this technique for the evaluation of muscle metabolism [19,20, 43–46]. To measure the absolute value, De Blasi *et al*. applied an arterial occlusion to calculate muscle oxygen consumption [47]. The application venous occlusion also allows the calculation of absolute values from NIRS data. Homma *et al*. also used venous occlusion method to measure the blood flow index and oxygen consumption index in forearm muscle during graded exercise [48].

In the field of peripheral vascular disease, NIRS has been used predominantly to investigate calf muscle oxygenation in arterial disease, and grades the intermittent claudication [10–13]. However, few studies have focused on the investigation of CVI using the NIRS. Hosoi *et al*. first applied this modality to assessing hemodynamics in patients with CVI. He used the ambulatory venous retension index (AVRI) obtained from the serial HHb changes as a parameter, and found that NIRS was useful in the evaluation of ambulatory venous dysfunction [14]. Later, he studied the symptomatic PTS, and found that the degree of AVRI was correlated well with clinical severity [15]. In the current study, we used the exercise protocol for APG because no standardized method for CVI is determined yet.

The main findings of this study using NIRS were that (i) HHbFI in POPV reflux patients was significantly higher than those with resolution; (ii) HHbRI in POPV reflux patients showed significantly higher value than those with resolution, and (iii) in patients with FV occlusion, the value of HHbRI was significantly higher than those with complete resolution. These results demonstrate that increased reflux in the POPV makes the major contribution to the calf muscle HHbFI. And increased HHbRI indicates the severity of both reflux and obstruction due to acute DVT. Previous reports support that increased reflux in the POPV shows an increasing severity of PTS, and residual obstruction in the major deep veins is associated with the development of the PTS [25,28,33]. This is also supported by AVRI obtained by NIRS which correlated well with the severity of PTS [15].

Our method has several advantages over recent diagnostic modalities in assessing PTS. First, NIRS provides non-invasive measurements of ambulatory venous hemodynamics using HHbFI and HHbRI. Despite the considerable incidence of PTS in clinical practice, objective assessment of the severity is still difficult. Duplex scanning is an essential tool for identifying the distribution and the extent of venous reflux as well as obstruction, but this useful tool does not reflect overall venous

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hemodynamics. Furthermore, duplex imaging is difficult to obtain under the condition of exercise, which can offer more physiologic information [15]. Secondly, venous occlusion does not affect NIRS-derived HHbFI and HHbRI. Data obtained by APG could evaluate ambulatory venous function, but APG parameters do not reflect correct data in the presence of proximal occlusion. Therefore, the NIRS-derived HHbFI and HHbRI could be promising parameters in the follow-up of acute DVT.

Our study also has a potential limitation. We do not correlate the NIRS with previously established methods for venous function for several reasons. The VCTs of >0.5 s are the gold standard for assessing the venous reflux, but recent studies have demonstrated that VCTs do not correlate with the severity of venous reflux nor should it be used as degree of reflux [49–55]. Calf muscle functions could be well evaluated using APG, but APG-derived parameters are not reliable when venous obstruction is present. Although validation study is required, we believe NIRS should work regardless of venous obstruction.

In conclusion, the lower extremity venous segments show different proportions of occlusion, partial recanalization, and total recanalization. The isolated CV shows more rapid recanalization than proximal veins. The NIRS-derived HHbFI and HHbRI could be promising parameters of the overall venous function in the follow-up of acute DVT. These findings might be very helpful to physicians in detecting patients who require much longer follow-up studies.

References

1 Franzeck UK, Schalch I, Jager KA, Schneider E, Grimm J, Bollinger A. Prospective 12-year follow-up study of clinical and hemodynamic sequelae after deep vein thrombosis in low-risk patients (Zurich study). *Circulation* 1996; **93**: 74–9.

2 Johnson BF, Manzo RA, Bergelin RO, Strandness DE Jr. The site of residual abnormalities in the leg veins in long-term follow-up after deep vein thrombosis and their relationship to the development of the post-thrombotic syndrome. *Int Angiol* 1996; **15**: 14–9.

3 Markel A, Manzo RA, Bergelin RO, Strandness DE Jr. Valvular reflux after deep vein thrombosis: incidence and time of occurrence. *J Vasc Surg* 1992; **15**: 377–84.

4 Janssen MCH, Wollersheim H, van Asten WNJC, de Rooij MJM, Novakova IRO, Thien T. The post-thrombotic syndrome: a review. *Phlebology* 1996; **11**: 86–94.

5 Haenen JH, Janssen MCH, van Langen H, van Asten WNJC, Wollersheim H, Heystraten FMJ, Skotnicki SH, Thien T. Duplex ultrasound in the hemodynamic evaluation of the late sequelae of deep venous thrombosis. *J Vasc Surg* 1998; **27**: 472–8.

6 Meissner MH, Caps MT, Zierler BK, Polissar N, Bergelin RO, Manzo RA, Strandness DE. Determinants of chronic venous disease after acute deep venous thrombosis. *J Vasc Surg* 1998; **28**: 826–33.

7 O'Shaughnessy AM, FitzGerald DE. The patterns and distribution of residual abnormalities between the individual proximal venous segments after an acute deep vein thrombosis. *J Vasc Surg* 2001; **33**: 379–84.

8 Shull KC, Nicolaides AN, Fernandes e Fernandes J, Miles C, Horner J, Needham T, Cooke ED, Eastcott FH. Significance of popliteal reflux in relation to ambulatory venous pressure and ulceration. *Arch Surg* 1979; **114**: 1304–6.

9 Beer A. Versus der Absorptions – Verhaltnisse des Cordietes fur rothes Licht zu bestimmen. *Ann Phys Chem* 1851; **84**: 37–52.

10 Cheatle TR, Potter LA, Cope M, Delpy DT, Coleridge-Smith PD, Scurr JH. Near-infrared spectroscopy in peripheral vascular disease. *Br J Surg* 1991; **78**: 405–8.

11 Komiyama T, Shigematsu H, Yasuhara H, Muto T. An objective assessment of intermittent claudication by near-infrared spectroscopy. *Eur J Vasc Surg* 1994; **8**: 294–6.

12 Komiyama T, Shigematsu H, Yasuhara H, Muto T. Near-infrared spectroscopy grades the severity of intermittent claudication in diabetics more accurately than ankle pressure measurement. *Br J Surg* 2000; **87**: 459–66.

13 Egun A, Fapooq V, Torella F, Cowley R, Thorniley MS, McCollum CN. The severity of muscle ischemia during intermittent claudication. *J Vasc Surg* 2002; **36**: 89–93.

14 Hosoi Y, Yasuhara H, Shigematsu H, Aramoto H, Komiyama T, Muto T. A new method for the assessment of venous insufficiency in primary varicose veins using near-infrared spectroscopy. *J Vasc Surg* 1997; **26**: 53–60.

15 Hosoi Y, Yasuhara H, Shigematsu H, Komiyama T, Onozuka A, Muto T. Influence of popliteal venous thrombosis on subsequent ambulatory venous function measured by near-infrared spectroscopy. *Am J Surg* 1999; **177**: 111–6.

16 Miller N, Satin R, Tousignant L, Sheiner MN. A prospective study comparing duplex scan and venography for the diagnosis of lower-extremity deep vein thrombosis. *Cardiovasc Surg* 1996; **4**: 505–8.

17 van Bemmelen PS, Bedford G, Beach K, Strandness DE. Quantitative segmental evaluation of venous valvular reflux with duplex ultrasound scanning. *J Vasc Surg* 1989; **10**: 425–31.

18 Delpy DT, Cope M, van der Zee P, Arrige S, Wray S, Wyatt J. Estimation of optical pathlength through tissue from direct time of light measurement. *Phys Med Biol* 1988; **33**: 1433–42.

19 Binzoni T, Quaresima V, Ferrari M, Hiltbrand E, Cerretelli P. Human calf microvascular compliance measured by near-infrared spectroscopy. *J Appl Physiol* 2000; **88**: 369–72.

20 Grassi B, Quaresima V, Marconi C, Ferrari M, Cerretelli P. Blood lactate accumulation and muscle deoxygenation during incremental exercise. *J Appl Physiol* 1999; **87**: 348–55.

21 Nicolaides AN, Christopoulos D. Quantification of venous reflux and outflow obstruction with air plethysmography. In: Bernstein EF, ed. *Vascular Diagnosis*. St Louis, MO, USA: Mosby, 1993: 915–21.

22 O'Shaughnessy AM, FitzGerald DE. Organization pattern of venous thrombus over time as demonstrated by duplex ultrasound. *J Vasc Invest* 1996; **2**: 75–81.

23 Caprini JA, Arcelus JI, Hoffman KN, Size G, Laubach M, Traverso CI, Coast R, Finke N, Reyna JJ. Venous duplex imaging follow-up of acute symptomatic deep vein thrombosis of the leg. *J Vasc Surg* 1995; **21**: 472–6.

24 McDonald PS, Kahn SR, Miller N, Obrand D. Short-term natural history of isolated gastrocnemius and soleal vein thrombosis. *J Vasc Surg* 2003; **37**: 523–7.

25 Killewich LA, Bedford GR, Beach KW, Strandness DE Jr. Spontaneous lysis of deep vein thrombosis: rate and outcome. *J Vasc Surg* 1989; **9**: 89–97.

26 Meissner MH, Manzo RA, Bergelin RO, Markel A, Strandness DE Jr. Deep veinous insufficiency: the relationship between lysis and subsequent reflux. *J Vasc Surg* 1993; **18**: 596–608.

27 Killewich LA, Martin R, Cramer M, Beach KW, Strandness DE. An objective assessment of the physiologic changes in the postthrombotic syndrome. *Arch Surg* 1985; **120**: 424–6.

28 Haenen JH, Janssen MCH, van Langen H, van Asten WNJC, Wollersheim H, van't Hof MA, Skotnicki SH, Thien T. The post-thrombotic syndrome in relation to venous hemodynamics, as measured by means of duplex scanning and strain-gauge plethysmography. *J Vasc Surg* 1999; **29**: 1071–6.

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29 Labropoulos N, Leon M, Nicolaides AN, Sowade O, Volteas N, Ortega F, Chan P. Venous reflux in patients with previous deep vein thrombosis: correlation with ulceration and other symptoms. *J Vasc Surg* 1994; **20**: 20–6.
30 Haenen JH, Janssen MCH, Wollersheim H, van't Hof MA, de Rooij MJM, van Langen H, Skotnicki SH, Thien T. The development of post-thrombotic syndrome in relation to venous reflux and calf muscle pump dysfunction at 2 years after the onset of deep vein thrombosis. *J Vasc Surg* 2002; **35**: 1184–9.
31 Nicolaides AN, Hussein MK, Szendro G, Christopoulos D, Vasdekis S, Clarke H. The relationship of venous ulceration with ambulatory venous pressure measurements. *J Vasc Surg* 1993; **17**: 414–9.
32 Harada RN, Katz ML, Comerota AJ. A noninvasive screening test to detect 'critical' deep venous reflux. *J Vasc Surg* 1995; **22**: 532–7.
33 Welkie JF, Comerota AJ, Katz ML, Aldridge SC, Kerr RP, White JV. Hemodynamic deterioration in chronic venous disease. *J Vasc Surg* 1992; **16**: 733–40.
34 Criado E, Farber MA, Marston WA, Daniel PF, Burnham CB, Keagy BA. The role of air plethysmography in the diagnosis of chronic venous insufficiency. *J Vasc Surg* 1998; **27**: 660–7.
35 Yoxell CW, Weindling AM. Measurement of venous oxyhaemoglobin saturation in the adult human forearm by near infrared spectroscopy with venous occlusion. *Med Biol Eng Comput* 1997; **35**: 331–6.
36 Jöbsis FF. Noninvasive infrared monitoring of cerebral and myocardial oxygen sufficiency and circulatory parameters. *Science* 1977; **198**: 1264–7.
37 Brazy JE, Lewis DV, Mitnick MH, Jöbsis van der Vliet FF. Noninvasive monitoring of cerebral oxygenation in preterm infants: preliminary observations. *Pediatrics* 1985; **75**: 217–25.
38 Edwards AE, Wyatt J, Richardson C, Delpy DT, Cope M, Reynolds EOR. Cotside measurement of cerebral blood flow in ill newborn infants by near infrared spectroscopy. *Lancet* 1988; **ii**: 770–1.
39 Wyatt JS, Cope M, Delpy DT, Richardson CE, Edwards AD, Wray S, Reynolds EOR. Quantification of cerebral blood volume in human infants by near infrared spectroscopy. *J Appl Physiol* 1990; **68**: 186–91.
40 Yoxall CW, Weindling AM, Dawani N, Peart I. Measurement of cerebral venous oxyhaemoglobin saturation in children with near infrared spectroscopy and partial jugular venous occlusion. *Pediatr Res* 1995; **38**: 319–23.
41 McCormick PW, Stewart M, Goetting MG, Balakrishnan G. Regional cerebrovascular oxygen saturation measured by optical spectroscopy in humans. *Stroke* 1991; **22**: 596–602.
42 Elwell CE, Cope M, Edwards AD, Wyatt JS, Reynolds EOR, Delpy DT. Measurement of cerebral blood flow in adult humans using near infrared spectroscopy – methodology and possible errors. *Adv Exp Med Biol* 1992; **317**: 235–45.
43 Chance B, Leigh J, Clark B, Maris J, Kent J, Nioka S. Control of oxidative metabolism and oxygen delivery in human skeletal muscle: a steady-state analysis of the work/energy cost transfer function. *Proc Natl Acad Sci USA* 1985; **82**: 8384–8.
44 Belardinelli R, Barstow T, Porszasz J, Wasserman K. Changes in skeletal muscle oxygenation during incremental exercise measured with near infrared spectroscopy. *Eur J Appl Physiol* 1995; **70**: 487–92.
45 De Blasi RA, Cope M, Elwell C, Safoue F, Ferrari M. Noninvasive measurement of human forearm oxygen consumption by near infrared spectroscopy. *Eur J Appl Physiol* 1993; **67**: 20–5.
46 Hamaoka T, Iwane H, Shimomitsu T, Katsumura T, Murase N, Nishio S, Osada T, Kurosawa Y, Chance B. Noninvasive measures of oxidative metabolism on working human muscles by near-infrared spectroscopy. *J Appl Physiol* 1996; **81**: 1410–7.
47 De Blasi RA, Ferrari M, Natali A, Conti G, Mega A, Gasparetto A. Noninvasive measurement of forearm blood flow and oxygenation consumption by near-infrared spectroscopy. *J Appl Physiol* 1994; **76**: 1388–93.
48 Homma S, Eda H, Ogasawara S, Kagaya A. Near infrared estimation of O_2 supply and consumption in forearm muscles working at varying intensity. *J Appl Physiol* 1996; **80**: 1270–84.
49 Yamaki T, Nozaki M, Sasaki K. Color duplex ultrasound in the assessment of primary venous leg ulceration. *Dermatol Surg* 1998; **24**: 1124–8.
50 Yamaki T, Nozaki M, Sasaki K. Predictive values in the progression of chronic venous insufficiency associated with superficial venous incompetence. *Int J Angiol* 2000; **9**: 95–8.
51 Yamaki T, Nozaki M, Sasaki K. Quantitative assessment of superficial venous insufficiency using duplex ultrasound and air plethysmography. *Dermatol Surg* 2000; **26**: 644–8.
52 Yamaki T, Nozaki M, Sasaki K. Preoperative duplex-derived parameters and angioscopic evidence of valvular incompetence associated with superficial venous insufficiency. *J Endovasc Ther* 2002; **9**: 229–33.
53 Yamaki T, Nozaki M, Fujiwara O, Yoshida E. Comparative evaluation of duplex-derived parameters in patients with chronic venous insufficiency: correlation with clinical manifestations. *J Am Coll Surg* 2002; **195**: 822–30.
54 Danielsson G, Grandinetti A, Lurie F, Kistner RL. Deep axial reflux, an important contributor to skin changes or ulceration in chronic venous insufficiency. *J Vasc Surg* 2003; **38**: 1336–41.
55 Neglén P, Egger JF III, Olivier J, Raju S. Hemodynamic and clinical impact of ultrasound-derived venous reflux parameters. *J Vasc Surg* 2004; **40**: 303–10.

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